

MIMIC-III ICU Operational Dashboard

Clinical analytics using admissions, ICU stays, LOS and mortality trends

47K

Total Patients

59K

Total Admissions

11.17

Avg Length of Stay

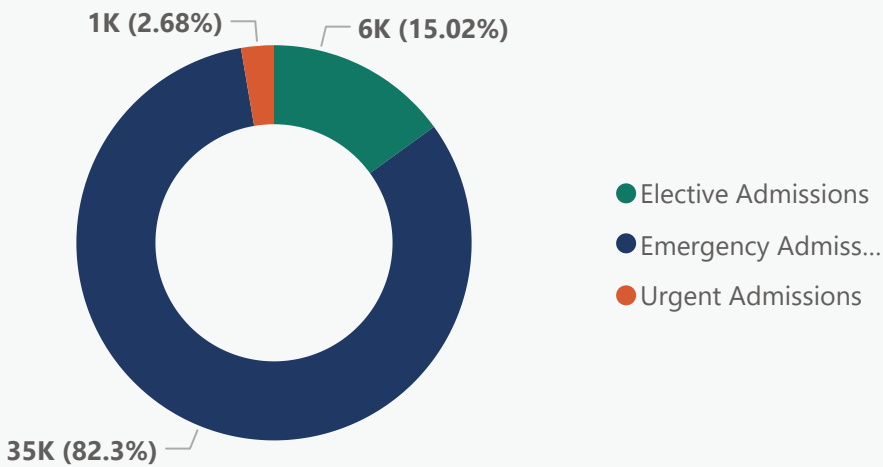
10.60

Mortality Rate %

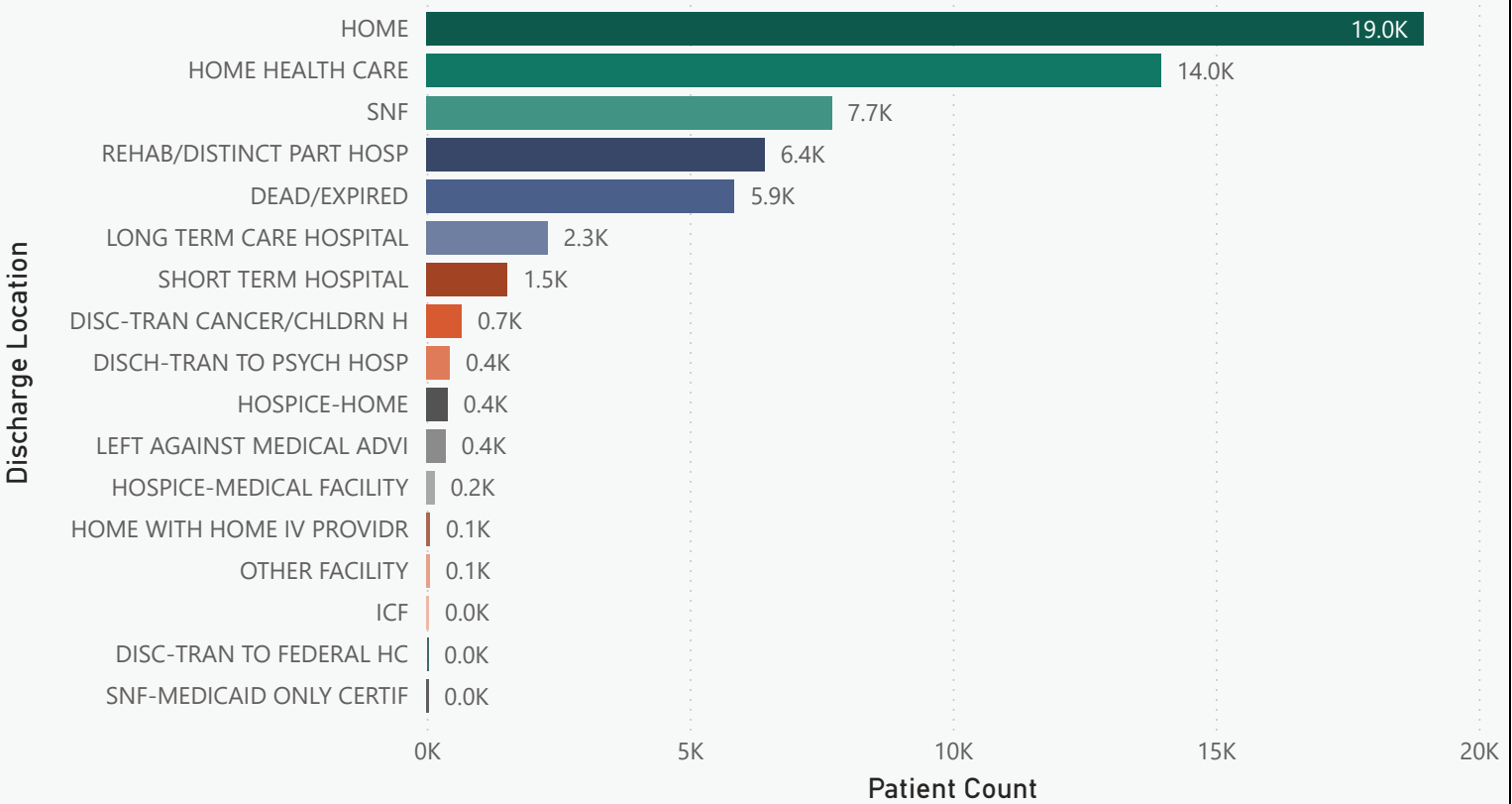
72.10

Emergency %

Admission Types

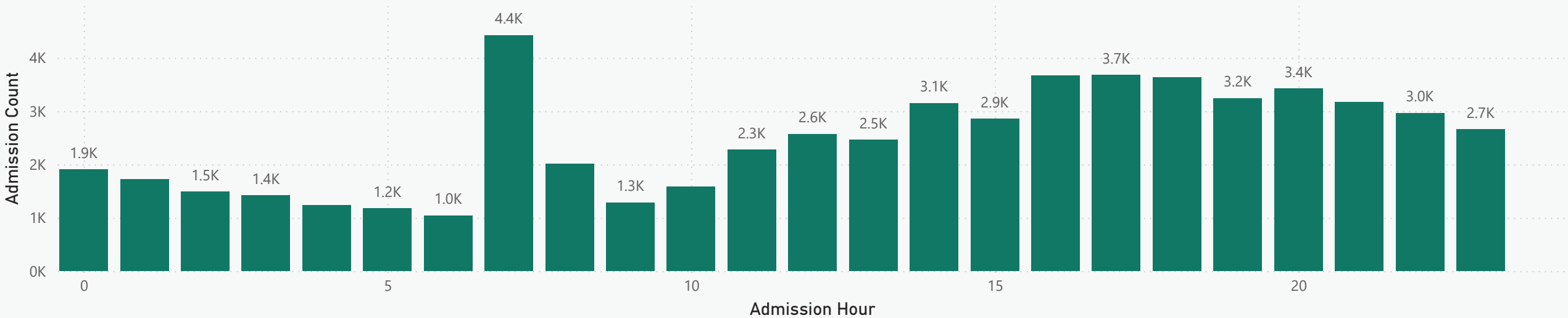


Discharge Location



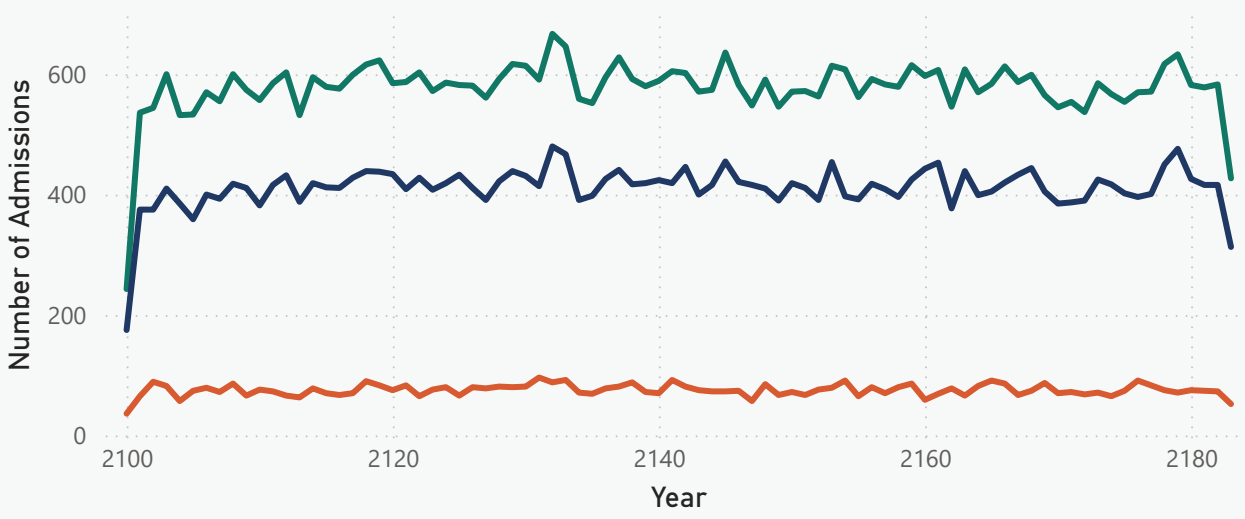
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Admissions by Hour of Day

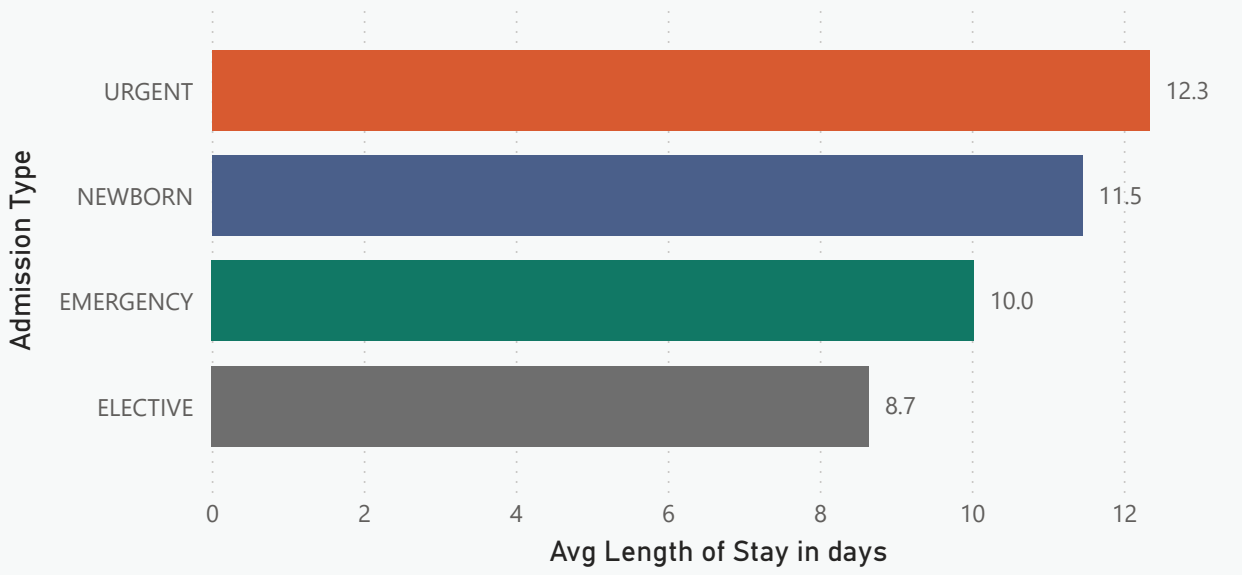


Monthly admissions over time - Emergency vs Elective

● Total admissions ● Emergency admissions ● Elective admissions



Avg LOS by admission type



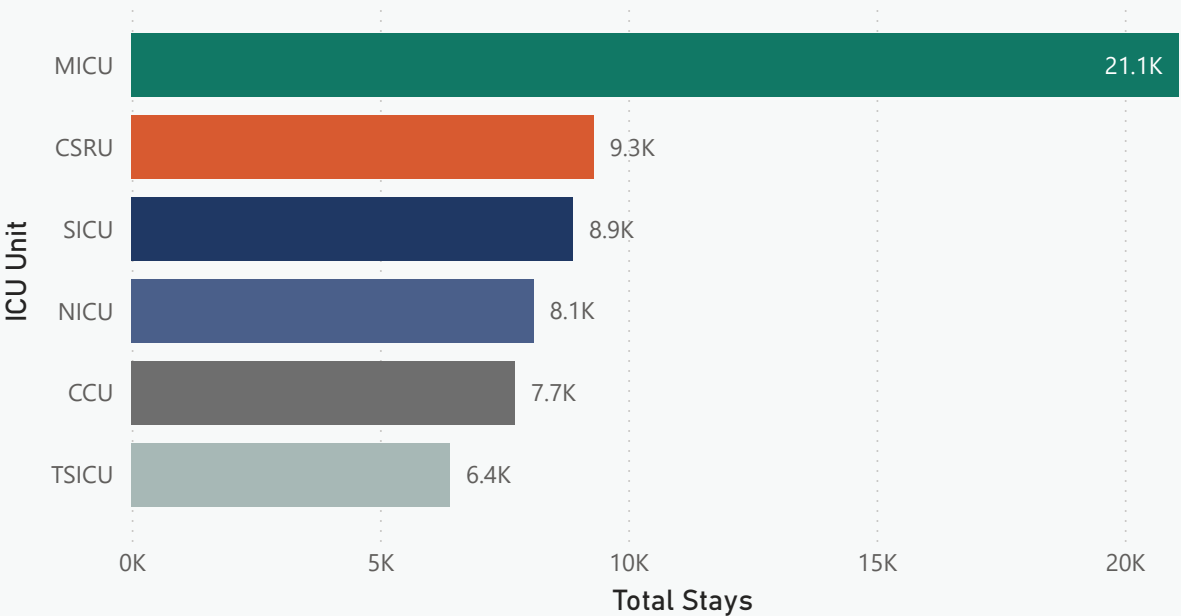
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ICU unit	Total stays	Avg los days	Min los days	Max los days	Std dev los	Prolonged stay %	Short stay %
CCU	7726	3.90	0	104.25	5.62	12.50	16.00
CSRU	9312	3.90	0	153.93	6.07	11.30	8.80
MICU	21088	4.01	0	116.83	5.84	13.50	16.80
NICU	8100	10.02	0	171.62	20.64	29.90	52.00
SICU	8891	4.71	0	101.74	6.94	17.40	15.30
TSICU	6415	4.44	0	173.07	6.65	16.50	17.90
Total	61532	30.98	0	821.44	51.76	101.10	126.80

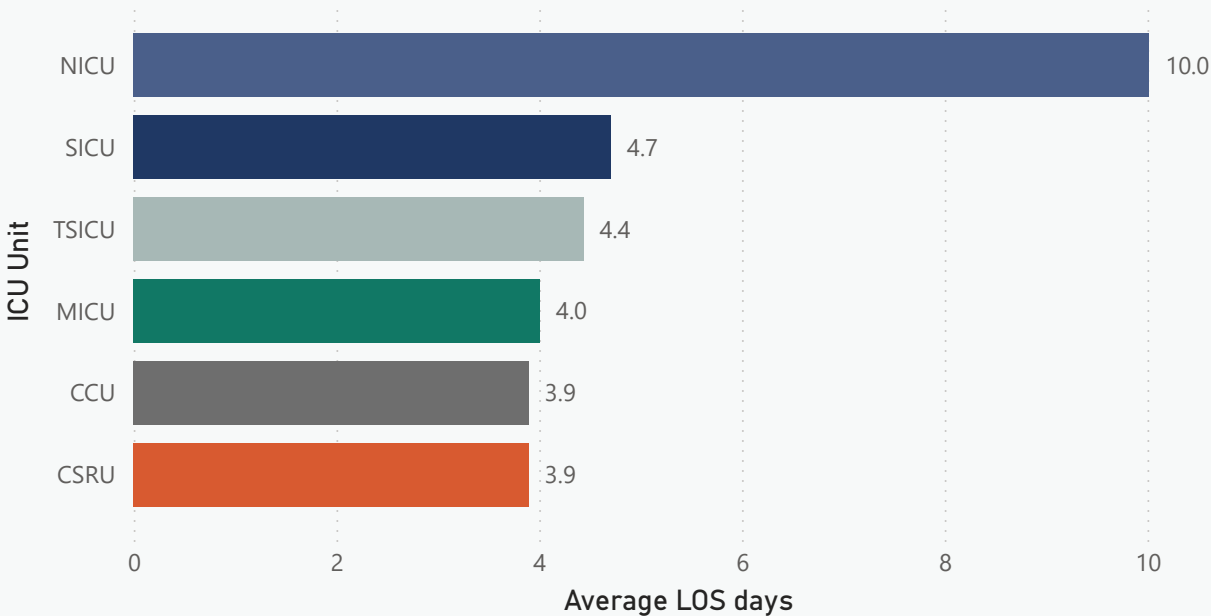
29.90

NICU Prolonged Stay %

Total Stays by ICU Unit

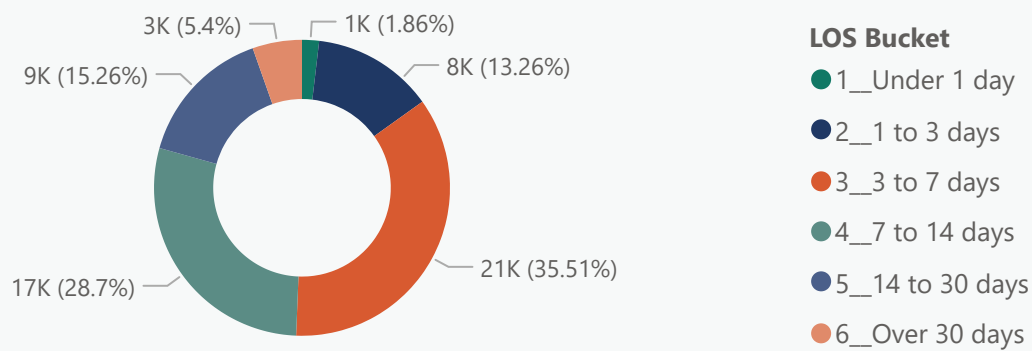


Average LOS days by ICU Unit

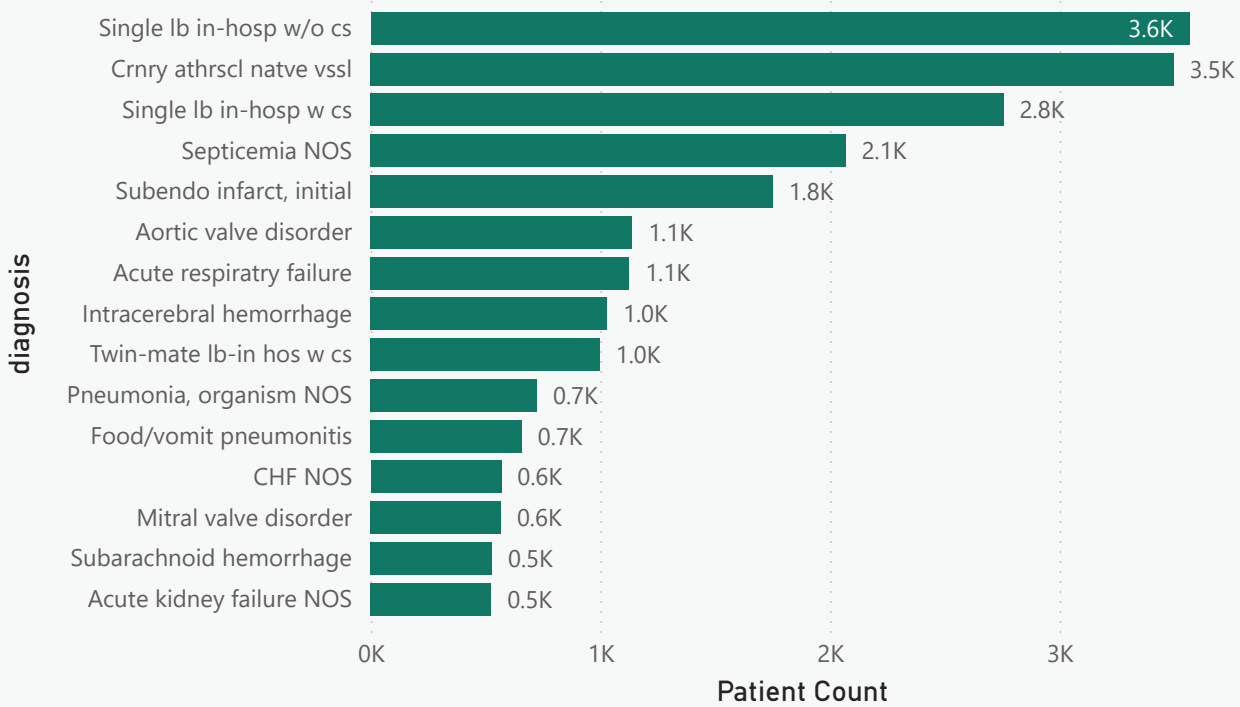


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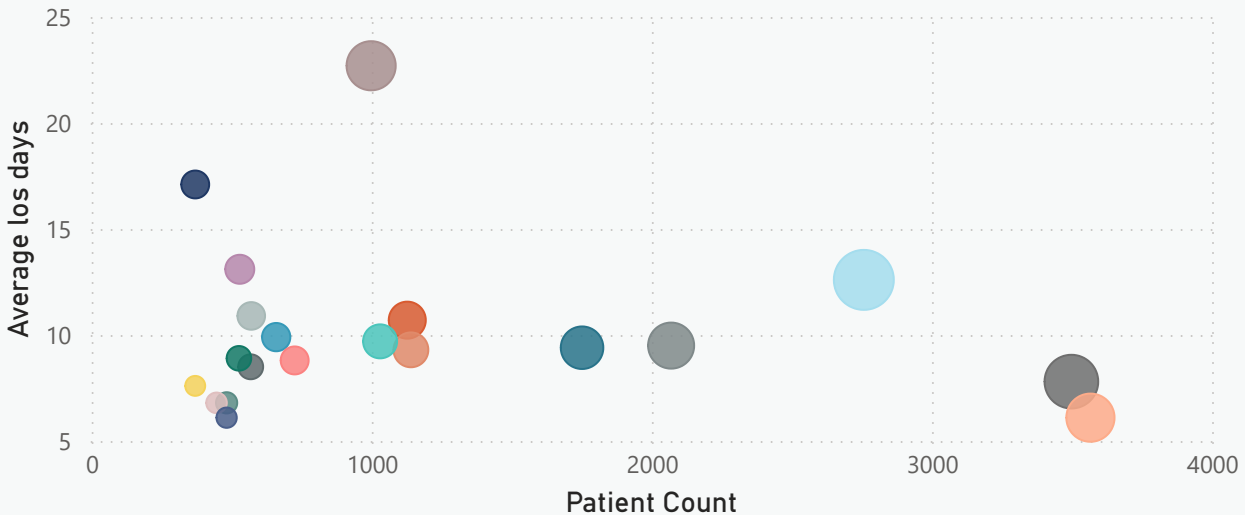
Patient Count by LOS Bucket



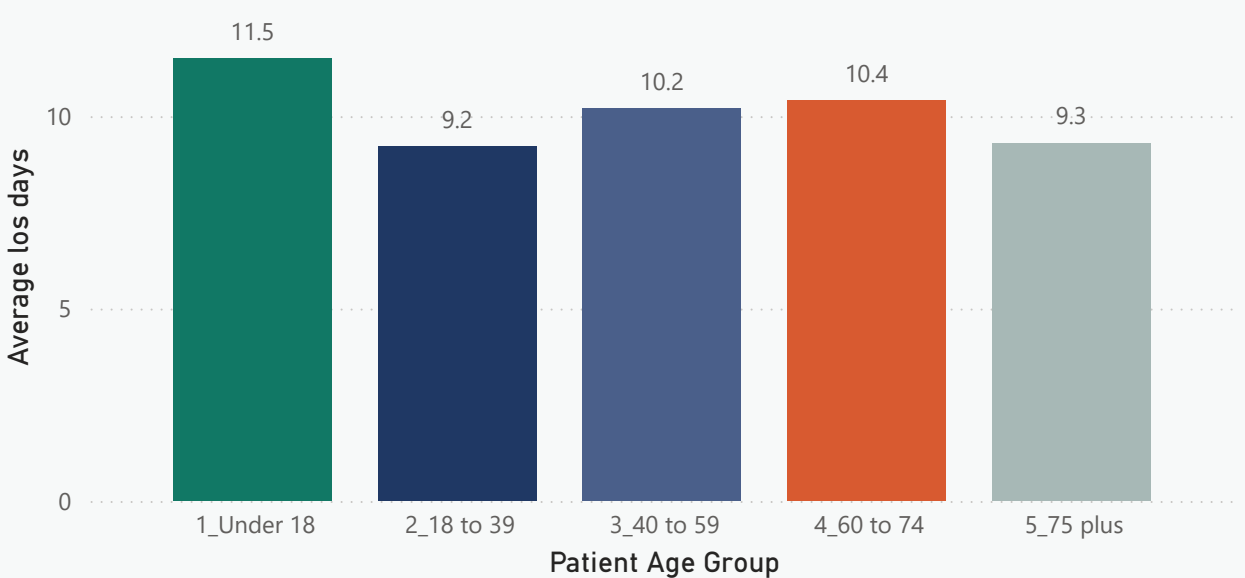
Patient Count by diagnosis



Common and Long-stay Diagnosis



Average los days by Patient Age Group



Operational Recommendations & Clinical Insights

Recommendations derived from ICU utilization, admission trends, and diagnosis burden analysis

ICU Capacity Optimisation

- MICU accounts for 34.3% of all ICU admissions and 13.5% of prolonged ICU stays (>7 days), indicating a disproportionately high operational burden on this unit. Targeted discharge planning protocols for MICU could reduce average ICU LOS by an estimated 1–2 days and improve bed availability during peak demand periods.

21K
MICU Admissions

Admission Flow Management

- Peak admission activity occurs during weekdays, particularly on Mondays, Tuesdays and Fridays. Current staffing likely understaffed for 7 AM when admission rate is highest. Introducing staggered shift start times could ease patient flow.

1138
Peak Hour Admission

Diagnosis-Specific LOS Reduction

- Top high-burden diagnosis groups including ‘Single liveborn in-hospital with/without cesarean section’, ‘Coronary atherosclerosis’, and ‘Septicemia NOS’ account for a substantial share of total hospital bed-days. Early specialist review protocol for these diagnoses could reduce LOS by 10–15% based on published operational benchmarks.

35K
Highest Bed-Day Burden